Kazakh-British Technical University

Data Collection & Preparation

STUDENTS' INDEPENDENT STUDY (SIS) No. 1

GDP per capita and life expectancy
across countries, 2023

Full Name: Yermyrzayev Aslan

ID: 23B031288

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GDP per capita and life expectancy across countries, 2023

Yermyrzayev Aslan | 23B031288 | Data Collection & Preparation | 12 October 2026

1. Question and data collection

This project examines whether higher GDP per capita is associated with longer life expectancy and compares the pattern across geographic regions. The unit of analysis is a country or area in 2023. A fixed year avoids combining observations from different years; 2023 is not presented as the latest available year.

The World Bank Indicators API supplies GDP per capita in current US dollars (NY.GDP.PCAP.CD) and life expectancy at birth in years (SP.DYN.LE00.IN) [1-3]. Country metadata identifies regional and income-group aggregates. The second source is the UN Statistics Division M49 webpage [4]. I extract its English HTML table with requests and BeautifulSoup, keeping ISO alpha-3 codes, names, regions, and subregions. The table is present in static HTML. The saved robots.txt permits access to the target URL [5].

Data were retrieved on 12 October 2026. Requests use timeouts, retries, pagination, and a one-second pause. Original JSON and HTML responses, full request URLs, access timestamps, and SHA-256 hashes are retained. Snapshot mode reproduces results without new requests. The UN classification reflects the access date, rather than a historical 2023 classification.

2. Cleaning and merging

The API produced 532 indicator records. Removing 98 aggregate records left 217 country/area rows after reshaping. The UN table contained 248 rows. Exact duplicates were checked (0 in the API table and 0 in the UN table). I standardized codes and column names, converted numeric fields, and checked unique country-year keys.

A validated one-to-one left merge uses ISO alpha-3 codes. There were 13 missing GDP values, 0 missing life-expectancy values, and 2 rows without a regional match. These reasons can overlap. In total, 15 distinct rows were excluded, leaving **202 complete observations**. Missing indicators were not imputed; valid extreme GDP values were retained. The notebook lists every exclusion and checks coverage by region. GDP is also transformed using log10 to reduce its right skew.

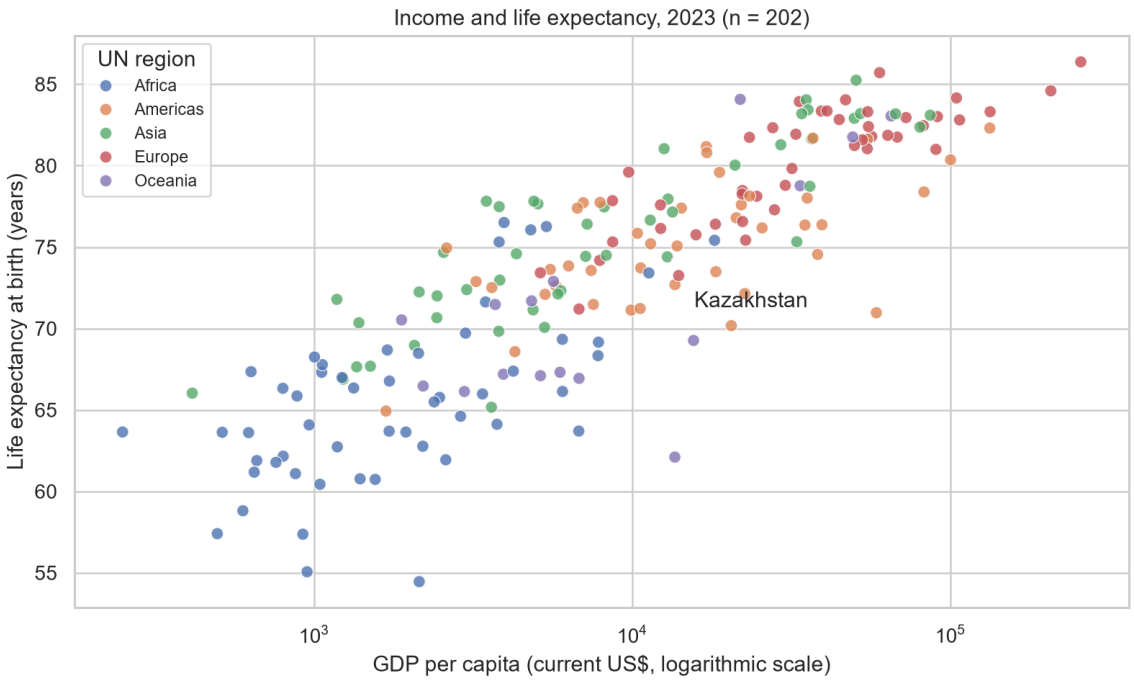


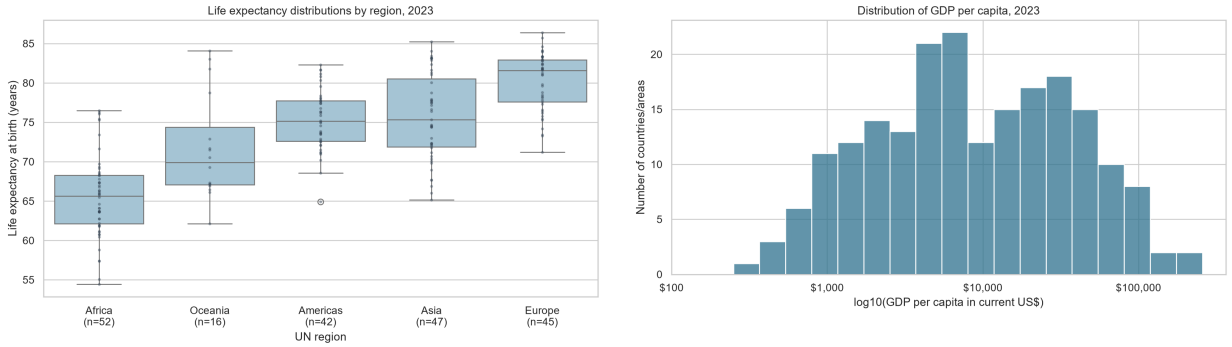
Figure 1. Countries/areas with higher GDP per capita generally have higher life expectancy. The GDP axis is logarithmic. Sources: World Bank WDI (2023 observations), UN M49 (accessed 12 October 2026).

3. Results

Across the complete sample, median GDP per capita is **US$7,862**, and median life expectancy is **74.5 years**. Pearson correlation is **0.620** for GDP on its original scale and **0.856** for log10 GDP. Spearman rank correlation is **0.866**. The stronger log-scale correlation is consistent with a curved relationship in the original units.

UN region	n	Median GDP (US$)	Median life (years)
Africa	52	1,709	65.6
Oceania	16	5,789	69.9
Americas	42	14,046	75.1
Asia	47	5,965	75.3
Europe	45	33,493	81.6

Kazakhstan has GDP per capita of US$12,879 and life expectancy of 74.4 years. A sensitivity check excluding the highest-GDP 5% still gives a positive log-GDP correlation of 0.844 (n = 191). All statistics weight countries/areas equally; regional summaries are not population-weighted estimates.



Figures 2-3. Regional life-expectancy distributions and the distribution of log10 GDP per capita. Full-size, labeled plots are included in the notebook and figures folder.

4. Interpretation and limitations

The data show a positive association between economic output per person and life expectancy, with substantial regional differences. This does not establish causation: health services, education, conflict, and inequality may influence both variables. GDP in current US dollars is sensitive to exchange rates and is not PPP-adjusted. Exclusions may bias coverage, countries differ in data quality, and a single year cannot show trends. A useful extension would compare PPP-adjusted GDP over several years.

References

[1] World Bank. GDP per capita (current US$), NY.GDP.PCAP.CD. Source link. Accessed 12 October 2026.

[2] World Bank. Life expectancy at birth, total (years), SP.DYN.LE00.IN. Source link. Accessed 12 October 2026.

[3] World Bank. API Basic Call Structures. Source link. Accessed 12 October 2026.

[4] United Nations Statistics Division. Standard country or area codes for statistical use (M49). Source link. Accessed 12 October 2026.

[5] United Nations Statistics Division. Robots rules. Source link. Accessed 12 October 2026.

AI assistance was used in preparing the code, analysis, and report. The assignment prohibits AI-generated submissions and sets 10 October 2026 as the submission deadline; these materials require instructor approval for assessed use.